Evidence-Based Multi-Layer Authorization for Voice Assistants: Calibrated Fusion and Cost-Derived Decision Thresholds

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Abstract—Voice assistants increasingly authorize payments and control physical devices, yet most deployed systems verify the speaker once, compare a similarity score against a hand-chosen cutoff, and treat every request as equally consequential. This paper presents SecureVoice, an authorization architecture that replaces hand-tuned score cutoffs with two principled mechanisms: per-layer probability calibration, which converts heterogeneous detector outputs into additive log-likelihood ratios, and a decision threshold derived from Bayes risk minimization, which rises logarithmically with the stake of the request. Four independent evidence layers—speaker verification, replay liveness, synthetic-speech detection, and a contextual layer that inspects the spoken command itself—feed a policy layer that returns allow, challenge, or deny. We report three findings from measurement rather than assertion. First, the customary naive-Bayes summation of layer scores is invalid in this setting because the layers test different hypotheses: an impostor speaking live is correctly reported as human by the anti-spoofing layer, and that correct statement cancels the correct rejection from the speaker layer, raising impostor false acceptance from 0.25% to 52.25%. Replacing summation with a conjunctive weakest-link rule removes the effect and guarantees that adding a layer can never open a hole. Second, a layer ablation shows every layer is necessary and none is redundant: removing liveness raises replay false acceptance from 1.25% to 74.0%, removing anti-spoofing raises clone false acceptance to 43.25%, and removing the contextual layer raises prompt-injection false acceptance from 10.0% to 23.5%. Third, a sensitivity sweep localizes the policy's dependence on its assumed constants almost entirely to the attack prior, which is the one constant a deployment can estimate from its own logs. We report all results with their scope, including two layers evaluated on small or synthetic corpora, and we identify these explicitly as requiring validation.

Index Terms—voice assistant security, speaker verification, anti-spoofing, replay detection, score calibration, log-likelihood ratio, Bayes risk, decision cost function, prompt injection

I. Introduction

Voice has become a transactional interface. Assistants embedded in phones, speakers, cars and appliances now place orders, move money and unlock doors, and the number of voice-enabled devices continues to grow. The security assumptions of these systems, however, were largely inherited from an era in which a voice command played music or reported the weather. When the worst outcome of a misrecognized command was an unwanted song, a permissive design was reasonable. When the worst outcome is an unauthorized payment, it is not.

Three properties of current deployments are, in our view, the source of the problem.

Authentication is a single event. A user is verified once, at the start of an interaction, and every subsequent command inherits that verification. An attacker who gains access mid-session inherits it too.

Verification reduces to one number against one cutoff. A speaker verification model emits a similarity score, the score is compared against a threshold, and the threshold is chosen by intuition or by tuning on a convenient dataset. The Phase-2 design that preceded this work followed exactly this pattern, with a cosine similarity of 0.75 treated as genuine, 0.85 as a strong match, and below 0.50 as an impostor. Nothing in that scheme explains why 0.75 rather than 0.70, and nothing connects the cutoff to what a wrong decision actually costs.

Every request is treated alike. A command that dims a lamp and a command that transfers a large sum are subjected to the same test. Where systems do distinguish between them, they typically do so through a small number of hand-drawn amount bands, which is a coarse approximation of a relationship that is, as we show, smooth and derivable.

These three properties interact badly with the modern threat landscape. Text-to-speech and voice-conversion systems now produce speech that speaker verification models accept as genuine [1], [2]. Replay attacks require no machine learning at all, only a recording and a loudspeaker. And once an assistant is mediated by a large language model, an attacker can address the model directly in speech, attempting to override its instructions—a voice-borne instance of the prompt injection problem documented in text-based LLM applications [4].

A. Contributions

This paper describes a system built to address those properties, and reports what measurement showed. Our contributions are:

1) Calibrated evidence in place of raw scores. Each layer's native output—a cosine similarity, a softmax

probability, a normality product—is mapped by logistic calibration onto a log-likelihood ratio, so that heterogeneous detectors become directly comparable and combinable (Section V-B).
2) A negative result on additive fusion. Calibrated LLRs are log-odds and therefore appear to justify summation. We show by measurement that summation is invalid here because the layers answer different questions, and that it degrades impostor detection from 0.25% to 52.25% false acceptance. We adopt a conjunctive weakest-link rule with the property that additional layers can only lower the fused score (Section V-C).
3) A threshold derived from cost, generalized beyond money. The decision boundary follows from Bayes risk minimization using a detection cost function [5], and rises logarithmically with the stake of the request. We generalize the monetary amount to a stake so that requests carrying no rupee figure, such as disclosing a one-time password, are not assigned the lowest possible bar (Section V-E).
4) A per-threat ablation over four layers and five threats, demonstrating that each layer covers a threat the others structurally cannot (Section VII-D).
5) A sensitivity analysis of the assumed cost constants, identifying which assumption the system's behavior actually depends on (Section VII-F).

B. Scope and honesty of reporting

Two of the four evidence layers rest on limited data, and we state this wherever their numbers appear rather than only in a closing paragraph. The liveness layer is fitted on 15 live/replay pairs from a single speaker, room and playback device. The contextual layer is calibrated on a template-generated command corpus, because no public corpus of labelled adversarial voice-assistant commands exists. Results derived from these are reported as indicative and are marked as requiring validation. No number in this paper is estimated, extrapolated or assumed; every figure is produced by a script in the accompanying implementation.

II. Literature Survey

A. Speaker verification

Modern text-independent speaker verification embeds an utterance into a fixed vector and compares vectors by cosine similarity. ECAPA-TDNN [6] refined the time-delay neural network family with channel attention, multi-layer feature aggregation and squeeze-excitation, and remains a strong and widely used baseline; we use the SpeechBrain implementation [7] trained on VoxCeleb [8]. Such models are highly effective against the threat they are designed for—a different human—and, importantly for this work, offer no defence at all against an accurate synthetic copy of the enrolled voice.

B. Spoofing countermeasures

The ASVspoof series established shared datasets and evaluation protocols for spoofing countermeasures [2], [3]. Logical access attacks comprise text-to-speech and voice conversion; physical access attacks comprise replay. Subsequent editions extended evaluation to compressed and transmitted speech and to deepfake audio [22]. Early countermeasures used engineered spectral features with generative classifiers; contemporary systems operate on the raw waveform. AASIST [9] applies a heterogeneous spectro-temporal graph attention network to a SincNet front end [10], and its lightweight variant reaches approximately 1% equal error rate on the ASVspoof 2019 logical access evaluation set with fewer than 10^5 parameters. Related end-to-end raw-waveform systems such as RawNet2 [23] follow the same trend away from hand-engineered front ends.

A recurring and well-documented weakness is generalization. Countermeasures trained on one corpus degrade substantially on unseen attacks and unseen recording conditions [11]. This is not a peripheral caveat; it is the dominant practical obstacle, and we encountered it directly (Section VII-B).

C. Liveness and replay detection

Replay is the cheapest attack on a voice interface and requires no synthesis. Detection exploits the fact that a replayed signal has passed through a loudspeaker and a second acoustic path, which reshapes its spectral envelope. Void [12] demonstrated that lightweight spectral features suffice for practical replay detection. A complementary line of work uses a second physiological channel: bone conduction captured by a wearable is present for genuine speech and absent for playback. That approach achieves high accuracy but requires hardware that a smart speaker does not have, which is why the present work pursues an air-conduction-only formulation. Vocal-tract-based methods have also been used to separate genuine from synthetic speech [13].

D. Attacks on the assistant itself

Beyond the acoustics of the speaker, the assistant is attackable as a system. Inaudible ultrasonic carriers can inject commands a user cannot hear [14]. Adversarial perturbations can cause a speech recognizer to transcribe an attacker's chosen text [15], and speaker recognition systems are themselves susceptible to adversarial examples [16]. Where a large language model interprets the command, the model's instruction-following behaviour becomes an attack surface: an attacker who controls part of the input can attempt to override the system prompt [4]. Surveys of voice assistant security consistently observe that individual defences exist for individual threats, while integrated frameworks do not [17].

TABLE I
Capabilities addressed in prior work and the integration gap

Capability	Representative work	Status
Speaker verification	ECAPA-TDNN [6]	Mature
Synthetic detection	AASIST [9]	Mature
Replay detection	Void [12]	Mature
Prompt injection	[4]	Emerging
Cost-based decisions	t-DCF [5]	Mature
Gap: calibrated fusion of the above into a single request-level authorization decision whose threshold is derived from the stake of that request rather than chosen by hand.		

TABLE II
Layers, the question each answers, and the threat each covers

Layer	Question	Threat covered
L1 Voiceprint	Is this the enrolled user?	Impostor (different human)
L2 Liveness	Mouth or loudspeaker?	Replay of a recording
L3 Anti-spoofing	Human or synthesizer?	TTS / voice conversion
L4 Context	Is this request normal?	Coerced or manipulated request
L5 Policy	Sets τ from the stake; decides	

E. Calibration and cost-based decisions

Detection scores are not probabilities, and comparing them across systems or against a cost requires calibration. Platt scaling [18] fits a logistic map from score to posterior. In speaker recognition, application-independent evaluation using calibrated log-likelihood ratios has been standard for two decades [19]. The ASVspoof challenges adopted a tandem detection cost function [5], [20] precisely because equal error rate ignores the operational cost of the two error types. Our threshold derivation follows directly from this literature; the contribution is its application to an amount-dependent authorization decision, and its generalization beyond monetary amounts.

III. Research Gap and Problem Statement

Table I summarizes the position. Each individual capability exists in the literature and is mature. What does not exist, to our knowledge, is a system that treats the outputs of several such capabilities as calibrated evidence for different hypotheses, fuses them under a rule that is correct for that structure, and compares the result against a threshold derived from the consequences of the specific request.

Problem statement. Given a spoken request to a voice assistant, decide whether to execute it, ask for further confirmation, or refuse it, such that (i) each defensive component contributes evidence on a common and interpretable scale, (ii) the combination rule is sound with respect to the distinct hypotheses the components test, (iii) the decision boundary is derived from the cost of being wrong about this particular request rather than selected by hand, and (iv) every element of the resulting decision is auditable.

IV. Proposed System and Methodology

A. Architecture

SecureVoice is organized as four independent evidence sources and one decision policy. We emphasize the distinction because it is frequently blurred: layers 1–4 are detectors that produce evidence; layer 5 is not a detector at all and produces no score. It sets the bar and makes the call.

The threat coverage in Table II is the substance of the layering argument. It is not that more layers are safer in some general sense; it is that each cell is a threat the other layers cannot see. An impostor is a real, live human, so L2 and L3 correctly pass them. A synthetic clone of the enrolled speaker is accepted by L1 precisely because the clone succeeded. A replayed recording of the genuine user is the genuine user's voice and a genuine human vocal tract, so both L1 and L3 pass it.

B. Threat model

We assume an attacker who can (i) speak near the device, (ii) obtain and replay a recording of the enrolled user, (iii) synthesize the enrolled user's voice from public samples using text-to-speech or voice conversion, and (iv) craft the linguistic content of a command freely, including content addressed to any language model in the pipeline. We assume the attacker cannot modify the device's software, extract stored embeddings, or alter the audit log. Attacks on the acoustic channel that bypass human hearing entirely, such as ultrasonic injection [14], and adversarial perturbations crafted against a specific model [15], [16], are out of scope for this evaluation, though we note that L4 operates on the transcript and is therefore unaffected by perturbations that leave the transcript intact.

C. Signal path

Audio is captured at 16 kHz. We note one preprocessing decision that proved consequential: browsers enable echo cancellation, noise suppression and automatic gain control by default, and these are aggressive spectral modifications. Since L3 asks whether a real vocal tract produced the waveform, supplying it with a signal that has already been spectrally reshaped is incorrect. Disabling this processing is required for correct operation (Section VII-B).

Transcription is performed locally with a Whisper-family model [21], so that audio never leaves the device; only the resulting text is available to downstream language-model components. This is a deliberate privacy property rather than an implementation convenience.

TABLE III
Fitted calibration parameters. Development data only; evaluation halves were held out and used once.

Layer	a_ℓ	b_ℓ	π_ℓ	n
L1 Voiceprint	6.176	−4.466	0.250	400
L2 Liveness	−2.869	1.428	0.500	30
L3 Anti-spoofing	−6.013	2.029	0.333	300
L4 Context	3.685	−0.802	0.508	120

V. Theoretical Foundation and System Architecture

A. Speaker verification and cosine similarity

ECAPA-TDNN maps a variable-length utterance to a fixed 192-dimensional embedding $\mathbf{e}$. Enrolment averages the embeddings of N enrolment utterances and renormalizes:

$$\mathbf{c} = \frac{\frac{1}{N}\sum_{i=1}^{N}\mathbf{e}_i}{\left\|\frac{1}{N}\sum_{i=1}^{N}\mathbf{e}_i\right\|_2}. \tag{1}$$

A test utterance with embedding $\mathbf{e}_t$ yields the L1 raw score

$$s_1 = \cos(\mathbf{c}, \mathbf{e}_t) = \frac{\mathbf{c}^\top \mathbf{e}_t}{\|\mathbf{c}\|_2 \, \|\mathbf{e}_t\|_2}. \tag{2}$$

We stress that s_1 is not a probability and is not comparable with the outputs of the other layers. Under the earlier tiered design, s_1 was compared directly against fixed cutoffs. This is the practice the present work replaces.

B. Calibration to log-likelihood ratios

Each layer's raw score s_ℓ is mapped to a posterior by a logistic model fitted on development data [18],

$$P(\text{genuine} \mid s_\ell) = \sigma(a_\ell s_\ell + b_\ell), \qquad \sigma(z) = \frac{1}{1+e^{-z}}, \tag{3}$$

and then converted to a log-likelihood ratio by removing the training prior π_ℓ of the calibration set:

$$\Lambda_\ell = \operatorname{logit} P(\text{genuine} \mid s_\ell) - \operatorname{logit} \pi_\ell. \tag{4}$$

Subtracting the prior is what makes the output evidence rather than a posterior contaminated by the class balance that the calibration set happened to have; our calibration sets are deliberately imbalanced, so the correction is not cosmetic. After Eq. (4) every layer speaks one language: $\Lambda_\ell > 0$ favours genuine by e^{Λ_ℓ} : 1 odds, $\Lambda_\ell = 0$ is uninformative, and $\Lambda_\ell < 0$ favours attack. Table III lists the fitted parameters.

C. Fusion: why summation fails

Because LLRs are log-odds, independent evidence for a single hypothesis combines by addition, and the naive-Bayes fusion $\Lambda = \sum_\ell w_\ell \Lambda_\ell$ is the obvious choice. It is also wrong here, and the reason is structural rather than numerical: addition presumes that all layers score the same hypothesis. Ours do not. For L1, "genuine" means the enrolled speaker; for L3 it means a real vocal tract.

Consider an impostor: a different human, speaking live. With our fitted calibrators, a similarity of 0.12 gives $\Lambda_1 = -2.63$, correctly rejecting. The same utterance has $P(\text{synthetic}) \approx 0.001$, giving $\Lambda_3 = +2.72$—also correct, because an impostor genuinely is a live human. Their sum is +0.09, which reads as genuine. A correct statement about one proposition has cancelled a correct statement about another, and the attacker is admitted.

This is not a hypothetical. Measured over 400 composite trials per cell, adding L3 to L1 under summation raises impostor false acceptance from 0.25% to 52.25%: a working layer made the system substantially worse.

The correct structure is conjunctive. Defence in depth means an attacker must defeat every layer, so a request is genuine only if every applicable layer finds it genuine. On log-odds this is the weakest link:

$$\Lambda \;=\; \min_{\ell \in \mathcal{A}} \; w_\ell \Lambda_\ell, \tag{5}$$

where $\mathcal{A}$ is the set of layers that produced a score. Weights w_ℓ default to unity and exist so that ablation can disable a layer without altering any other code path. Eq. (5) has the property the security argument requires: adding a layer can only decrease Λ, so a new layer can never open a hole. It can only cost false rejections, and that is precisely the trade our ablation measures.

D. The contextual layer

L1 to L3 all interrogate the signal. None can observe the request. If an attacker has the genuine user present and coerced, or speaks an instruction-override attack into the assistant, the audio is authentically the correct human, genuinely alive, and L1 to L3 correctly pass it. Only a layer that reads the content and the user's history can object.

L4 combines three indicators, each in $[0, 1]$ and each meaning "nothing wrong here": an injection probability from a dedicated classifier; a social-engineering pressure score derived from language-model-extracted markers (urgency, secrecy, claimed authority, threatened consequence), taken as the maximum rather than the mean, since a request that is purely an authority claim is not made harmless by lacking urgency; and a behavioural comparison of the request against the user's own transaction history in log-amount space. The three are multiplied, so one strong indicator dominates, and an unavailable indicator is omitted rather than imputed.

Table IV reports each indicator alone on held-out attacks. The result argues against relying on any single one: the dedicated injection classifier is excellent on literal instruction-override phrasing but detects only a quarter of attacks overall, while the semantic markers cover most of the remainder. This is the layering argument reproduced inside a single layer.

Calibrated on 120 development commands and evaluated once on 120 held out, L4 reached 16.67% development

TABLE IV
L4 sub-indicators alone, on held-out attack commands ($n = 61$). Corpus is template-generated; see limitations.

Indicator alone	Mean score	Attacks caught
Injection classifier	0.720	24.6%
Semantic markers	0.183	91.7%
Behavioural fit	0.490	67.2%
Product of all three	—	85.8% acc.

EER and 13.34% evaluation EER, with per-family accuracy of 89.8% on genuine commands, 92.3% on injection and 74.3% on social engineering. These figures describe separation on a template corpus and are not deployment estimates.

E. Cost-derived thresholds

A detector should minimize expected cost rather than maximize accuracy. For a binary decision with miss and false-alarm costs $C_{\text{miss}}, C_{\text{fa}}$ and target prior P_{tar}, the expected cost is

$$E[C] = C_{\text{miss}} P_{\text{miss}} P_{\text{tar}} + C_{\text{fa}} P_{\text{fa}} (1 - P_{\text{tar}}), \tag{6}$$

which is the form adopted by the ASVspoof detection cost function [5]. Minimizing it yields the Bayes decision rule: accept when the calibrated LLR exceeds

$$\tau = \log \frac{C_{\text{fa}}}{C_{\text{miss}}} + \log \frac{\pi}{1 - \pi}, \tag{7}$$

where π is the prior probability that a request is an attack.

The two error costs in an authorization setting are asymmetric, and only one of them scales. Wrongly approving a fraudulent request costs what was taken, plus a fixed handling cost: $C_{\text{fa}}(A) = A + C_h$. Wrongly refusing a genuine user costs a support interaction and some goodwill, and does not depend on A. Substituting gives an amount-dependent threshold that rises logarithmically, which is a derived curve rather than a set of hand-drawn tiers.

We use three outcomes rather than two, because the most useful response to inconclusive evidence is to ask again. Two boundaries follow from two different costs, with $C_{\text{chal}} < C_{\text{rej}}$:

$$\tau_{\text{allow}}(A) = \log \frac{A + C_h}{C_{\text{chal}}} + \log \frac{\pi}{1 - \pi}, \tag{8}$$

$$\tau_{\text{deny}}(A) = \log \frac{A + C_h}{C_{\text{rej}}} + \log \frac{\pi}{1 - \pi}. \tag{9}$$

Between them lies the challenge band (Fig. 1).

F. Generalizing amount to stake

An amount-only policy has a defect that is easy to miss: a request carrying no monetary figure yields $A = 0$ and therefore the lowest threshold the system can issue. "Forward the one-time password" and "unlock the front door" would receive the cheapest verification in the system.

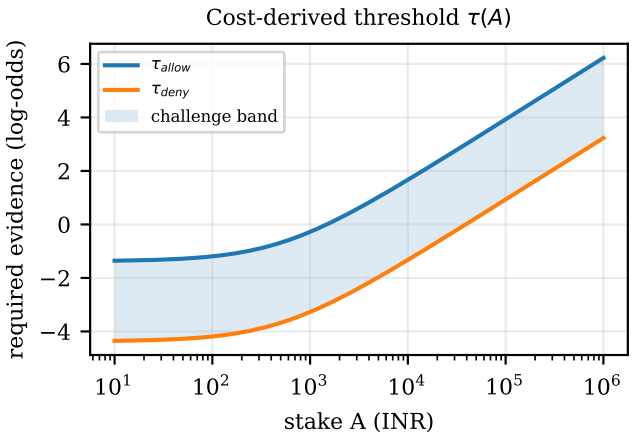


Fig. 1. Decision thresholds derived from Eqs. (8) and (9). The required evidence grows logarithmically with the stake; the shaded region is where the system asks for re-verification instead of committing.

The Bayes rule needs C_{fa}, the cost of wrongly approving. A rupee amount is one way to estimate that quantity, not its definition. We therefore keep the derivation untouched and broaden the estimate:

$$A = \max\big(A_{\text{spoken}},\ A_{\text{consequence}},\ A_{\text{price}}\big), \tag{10}$$

where $A_{\text{consequence}}$ is an anchor for the class of action and A_{price} is an estimate of the price of a named item. Because the combination is a maximum, an underestimate—the dangerous direction, since τ increases with A—is absorbed by the anchor and changes nothing, while an overestimate costs only friction. A fallible estimator can therefore inconvenience a user but cannot open a hole.

The action class is assigned by a language model constrained to a fixed enumeration and validated against it. The model never emits a currency figure and never sees the anchor table; if it could name its own stake, an attacker controlling the transcript would control the threshold. The anchors themselves are declared policy constants, not measurements, and we treat them as assumptions throughout.

Table V shows the effect. A request that toggles a lamp and a request that discloses a one-time password differ by roughly seven units of log-odds in the evidence they must clear, and both figures come from the same formula.

G. The assistant as an attack surface

Where a language model mediates the interaction, it is an attack surface rather than an authority. We enforce one invariant: no language model output is ever a verdict. The model may emit features, which enter the decision only after passing through the same calibration of Eq. (4) as every other layer, and it may verbalize a decision that has already been made. It cannot produce an outcome, a threshold or an LLR. Every spoken response

TABLE V
Stake and resulting evidence requirement. Anchors are declared policy constants, not measurements.

Request	Stake (INR)	τ_{allow}
"What is the weather today?"	0	−1.38
"Turn off the lights."	100	−1.19
"Order a packet of biscuits."	1 000	−0.28
"Transfer 10 000 rupees."	10 000	1.67
"Buy a new laptop."	350 000	5.18
"Forward the one-time password."	500 000	5.53

TABLE VI
Per-layer calibrated evidence (LLR) for one utterance, spoken live and replayed. L1 and L3 pass the replay, correctly.

Layer	Live	Replayed	Reading
L1 Voiceprint	+0.48	+0.21	correct voice in both
L2 Liveness	+1.37	**−1.35**	only layer that objects
L3 Anti-spoofing	+2.72	+2.70	real vocal tract in both
L4 Context	+1.93	+2.11	same benign request
Fused (min)	+0.48	−1.35	
Outcome	ALLOW	CHALLENGE	

has a deterministic template fallback, and a response contradicting the computed decision is discarded in favour of that template.

H. Decision procedure

Input: audio x, enrolled centroid $\mathbf{c}$, history H
$T \leftarrow \text{ASR}(x)$ {local, on-device}
$s_1 \leftarrow \cos(\mathbf{c}, \text{embed}(x))$
$s_2 \leftarrow P(\text{playback} \mid x)$; $s_3 \leftarrow P(\text{synthetic} \mid x)$
$s_4 \leftarrow \text{context}(T, H)$
$A \leftarrow$ stake from T by Eq. (10)
for $\ell \in \{1, 2, 3, 4\}$ with a score do
    $\Lambda_\ell \leftarrow \sigma(a_\ell s_\ell + b_\ell)$ mapped by Eq. (4)
end for
$\Lambda \leftarrow \min_\ell w_\ell \Lambda_\ell$
compute $\tau_{\text{allow}}(A), \tau_{\text{deny}}(A)$
if $\Lambda \geq \tau_{\text{allow}}$ then
    return ALLOW
else if $\Lambda \geq \tau_{\text{deny}}$ then
    return CHALLENGE
else
    return DENY
end if

A layer that produces no score is reported unavailable rather than silently skipped or imputed as neutral, so that an audit record states which defences were actually in play.

VI. Implementation Details

The prototype is a FastAPI service with a browser front end. L1 uses the SpeechBrain ECAPA-TDNN model [7] trained on VoxCeleb. L3 uses AASIST-L [9], a raw-waveform graph attention network of approximately 8.5×10^4 parameters operating on a fixed 64 600-sample (4.04 s) window. L2 is a logistic model over spectral and cepstral summary features. L4 combines an injection classifier, a language-model extraction of social-engineering markers, and a behavioural comparison against the user's own history; the three are multiplied as independent "nothing wrong here" indicators.

Every authorization writes an audit record containing the user identifier, timestamp, transcript, each layer's raw score and calibrated LLR, the derived thresholds, the stake and its derivation, and the outcome. Because the threshold is computed rather than looked up, the audit record is sufficient to reconstruct any decision.

The CHALLENGE outcome is actioned rather than merely reported: the assistant speaks, requests the enrolled passphrase, and the second utterance is scored by the same layers. Text verification is a precondition—if the phrase does not match, the challenge fails regardless of voice quality.

VII. Results and Discussion

A. Component baselines

On ASVspoof 2019 logical access, AASIST-L reproduced its published performance under our pipeline: 0.51% equal error rate on the development set (24 844 utterances) and **0.99**% on the evaluation set (71 237 utterances), at approximately 67 ms per utterance. An LFCC-GMM baseline implemented for comparison reached 9.38% development and 18.75% evaluation EER. These are in-domain results and should be read as such.

A tandem experiment over 800 trials quantified the gap that motivates L3: the speaker verification layer rejected 100% of impostors but admitted 54.0% of synthetic clones, of which L3 subsequently caught 87.0% with no false alarms on genuine trials.

B. The domain gap

Anti-spoofing performance is capture-dependent, and we observed this severely. Genuine speech recorded through the browser was initially assigned $P(\text{synthetic}) = 1.0$—the model's maximum confidence—while the same speaker recorded on a phone scored 0.000. The cause was the browser's default signal processing. With echo cancellation, noise suppression and automatic gain control disabled, false positives on 20 genuine browser-captured clips fell to 1/20, better than the phone reference (2/8). We report this because it is an easily reproduced failure with an easily missed cause, and because it supports a general rule: forensic audio layers must receive unprocessed capture.

C. Case study: the same utterance, spoken and replayed

Table VI reports the per-layer evidence for one utterance spoken live and then replayed through a loudspeaker and re-recorded on the same microphone. It illustrates the argument for layering more directly than an aggregate table can.

TABLE VII
False acceptance rate (%) by threat, conjunctive fusion. Last column is false rejection of genuine users. Bold marks the blindness created by removing a layer.

Subset	Imp.	Replay	Clone	Inj.	Soc.	FRR
L1 only	0.25	74.00	43.25	23.50	19.50	26.00
L1+L3	0.25	74.00	1.50	23.50	19.50	26.00
L1+L2+L3	0.25	1.25	1.50	23.50	19.50	27.00
L1+L3+L4	0.25	62.00	1.25	10.00	7.25	38.00
All four	0.25	1.25	1.25	10.00	7.25	38.75

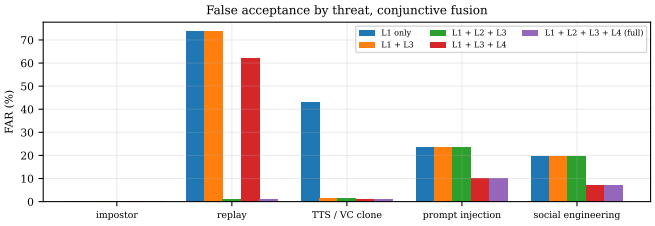


Fig. 2. False acceptance by threat for each layer subset. Every subset is blind to at least one threat; the full system is best or tied-best in every column.

L1 and L3 do not fail on the replay; they succeed. The utterance genuinely is the enrolled user's voice, and it genuinely was produced by a human vocal tract. Under an additive rule their two correct positive statements would outweigh L2's single correct negative one; under Eq. (5) the objection stands.

D. Layer ablation

Table VII and Fig. 2 report false acceptance by threat for layer subsets, over 400 composite trials per cell on held-out data. Trials pair independently measured audio, liveness and command conditions; this evaluates the fusion policy and assumes layer errors are independent, an assumption a real deployment may violate.

Every subset is blind to something, and the blindness is structural rather than statistical. Removing L2 raises replay false acceptance from 1.25% to 74.0%, and the reason is visible in the per-layer evidence: for a replayed utterance L1 and L3 both pass, correctly, because it is genuinely the enrolled user's voice and genuinely a human vocal tract. Only L2 objects.

The cost of the conjunctive rule is false rejection, which rises from 26.0% to 38.75% as layers are added. This figure should not be read as customers turned away: of genuine users, 61.25% are allowed immediately, 26.75% are challenged and recoverable through the spoken passphrase loop, and 12.0% are refused outright. This is what the three-outcome policy purchases, and it is why $C_{\text{chal}} \ll C_{\text{rej}}$ in the cost model.

E. Fusion rule comparison

Fig. 3 contrasts the two rules on the full system. Under summation the ordering of subsets is non-monotonic and

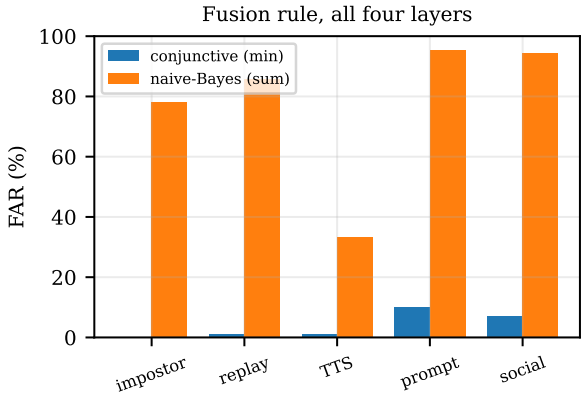


Fig. 3. Conjunctive versus additive fusion, all four layers. Summation is invalid here because the layers test different hypotheses.

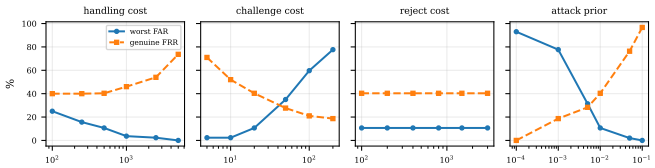


Fig. 4. Sensitivity of worst-case false acceptance and genuine false rejection to each assumed constant. Dependence is concentrated in the attack prior.

impostor false acceptance is an order of magnitude worse. We report this as a negative result about a standard technique applied to a non-standard structure.

F. Sensitivity to the assumed constants

The threshold's form is derived, but its four constants $(C_h, C_{\text{chal}}, C_{\text{rej}}, \pi)$ are assumptions, since we have no fraud logs from which to estimate them. The material question is therefore not whether they are correct but how wrong they can be before behaviour changes. Fig. 4 sweeps each over at least an order of magnitude.

The result is informative and partly uncomfortable. Varying C_{rej} changes which requests are challenged versus denied but leaves both FAR and FRR unchanged, because it moves only the lower boundary. C_h and C_{chal} behave as smooth, monotonic security-versus-friction dials. The attack prior π, however, dominates: across the swept range, worst-case FAR moves between 0.0% and 93.0% and genuine FRR between 0.0% and 96.7%.

We regard this as a useful localization rather than a refutation. Three of the four constants are business policy and behave benignly; the one constant that matters is the observed fraud rate, which is exactly the quantity a deployment can estimate from its own logs. Our default $\pi = 0.01$ is conservative relative to typical transaction fraud rates, so error in this assumption biases the system toward more verification rather than less.

TABLE VIII
Design comparison. The right column's parameters are auditable quantities rather than score cutoffs.

	Tiered baseline	This work
Decision input	one cosine similarity	four calibrated LLRs
Threshold	fixed cutoffs (0.75 / 0.85)	derived, $\tau(A)$
Amount handling	3–4 hand-drawn bands	continuous, logarithmic
Non-monetary risk	not represented	stake, Eq. (10)
Tunable numbers	score cutoffs	four business costs
Threats covered	impostor	impostor, replay, clone, manipulated request
Auditability	score and verdict	full derivation

G. Liveness layer

Under leave-one-pair-out cross-validation over 15 pairs, L2 achieved 100% accuracy and 0% EER. An identical figure from an earlier iteration of this project proved to be an artefact—live clips were recorded on one device and replays on another, so the classifier separated devices. The present corpus was captured so that each pair is recorded through one microphone seconds apart, clips are cropped to a common length to remove a duration cue, and 20 further genuine clips, unseen in training, were classified live at 20/20. The discriminating features are low-order cepstral means, consistent with the spectral reshaping expected from an additional loudspeaker and room path. This result nevertheless rests on one speaker, one room and one playback device—a laptop's built-in speakers, which are poor reproducers—and must be validated against high-fidelity playback before it can be described as a replay detector in general.

H. Comparison with the tiered baseline

Table VIII contrasts the present design with the tiered similarity-threshold scheme it replaces. The distinction we wish to draw is not that one set of numbers is larger than another, but that the quantities in the right-hand column are answerable: each is either measured, or is a stated business assumption that a reviewer can dispute specifically.

VIII. Security, Privacy, and Limitations

A. Security properties

The conjunctive rule of Eq. (5) gives a monotonicity guarantee: adding a layer cannot increase Λ, so no new layer can admit an attack that the previous configuration refused. Deriving the stake from speech does not give an attacker leverage, because the stake sets only strictness; an attacker who understates it to obtain a lower bar thereby restricts themselves to a low-stake action. The language model's exclusion from the verdict path means that a successful prompt injection can at most corrupt a feature, which is then bounded by calibration, rather than dictate an outcome.

B. Privacy

Speech is biometric data. Transcription runs locally, and only text is exposed to any external model, so raw audio does not leave the device. Voiceprints are stored as 192-dimensional embeddings rather than audio. We note honestly that embeddings are not anonymous—they identify the speaker by construction—and that a deployment must treat them as sensitive personal data under the applicable regime.

C. Limitations

1) L4's corpus is template-generated. No public corpus of labelled adversarial voice-assistant commands exists. Reporting separation on this corpus is legitimate; reporting its EER as a deployment estimate is not.
2) L2 rests on 15 pairs, one speaker, one room, one playback device, as stated above.
3) Composite trials assume independent layer errors. No corpus carries all conditions on one utterance. Correlated failures are plausible in deployment.
4) Live testing used a single speaker. False rejection estimates from it do not generalize; demographic bias is consequently unmeasured, and speaker verification is known to exhibit performance disparities across demographic groups.
5) The stake anchors and cost constants are assumptions, and the action category is assigned by a language model, so category errors propagate to the threshold.
6) The passphrase is fixed, so any recording of the user speaking it satisfies the text check. A randomly chosen challenge phrase would remove this weakness and is the natural next step.
7) Voice conversion is under-tested relative to text-to-speech, and prior work reports that it is the harder attack family.

IX. Conclusion and Future Work

We presented an authorization architecture for voice assistants in which heterogeneous detectors are calibrated to a common evidence scale, fused under a rule appropriate to the fact that they test different hypotheses, and compared against a threshold derived from the cost of being wrong about the specific request. Measurement, rather than argument, established three results: that additive fusion of calibrated scores is invalid in this setting and degrades impostor detection by two orders of magnitude; that each of the four evidence layers covers a threat the others structurally cannot, so that no layer is redundant; and that the policy's dependence on its assumed constants is concentrated almost entirely in the attack prior, which is the one constant a deployment can measure.

Future work follows directly from the limitations. A randomly chosen challenge phrase would give replay resistance independent of the acoustic liveness layer. A larger and more diverse liveness corpus, including high-fidelity playback and multiple speakers and rooms, is required before the liveness result can be generalized. Collecting genuine adversarial commands, rather than templates, would put L4 on the same evidentiary footing as L1 and L3. Estimating the attack prior from deployment logs would convert the system's most sensitive assumption into a measurement. Finally, a demographic evaluation is necessary before any claim of fairness can be made.

Reproducibility

All reported numbers are produced by scripts in the accompanying implementation. Calibration uses development splits only; evaluation halves were scored once. Random seeds are fixed and recorded alongside every result file.

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